

ASSIGNMENT 6

Selenium Web Basics – 1

Q1. Explain WebDriver Architecture in Selenium

WebDriver Architecture:

Selenium WebDriver follows a client-server architecture.

1. Selenium WebDriver script is written in Java.
2. WebDriver sends commands to the browser driver (ChromeDriver, GeckoDriver, EdgeDriver).
3. The browser driver converts commands into browser-understandable instructions.
4. The browser performs the action and sends the response back.

Diagram

Java Selenium Script

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v

WebDriver

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v

ChromeDriver

Chrome Browser

Q2. Selenium Program to Search in Google

```
import org.openqa.selenium.By;  
import org.openqa.selenium.WebDriver;  
import org.openqa.selenium.chrome.ChromeDriver;
```

```
public class GoogleSearch {  
  
    public static void main(String[] args) {  
  
        WebDriver driver = new ChromeDriver();  
  
        driver.get("https://www.google.com");
```

```

        driver.findElement(By.name("q"))
            .sendKeys("Selenium WebDriver");

        driver.findElement(By.name("q"))
            .submit();
    }
}

```

Q3. Different Locators in Selenium

Locators are used to identify web elements.

Locator	Description	Example
id	Finds element using id attribute	<code>By.id("username")</code>
name	Finds element using name attribute	<code>By.name("email")</code>
className	Finds element using class name	<code>By.className("btn")</code>
tagName	Finds element using HTML tag	<code>By.tagName("input")</code>
linkText	Finds link using exact text	<code>By.linkText("Login")</code>

partialLinkText	Finds link using partial text	By.partialLinkText("Log")
xpath	Finds element using XPath	By.xpath("//input[@id='user']")
cssSelector	Finds element using CSS selector	By.cssSelector("#user")

Example

```
driver.findElement(By.id("username"));
driver.findElement(By.name("email"));
driver.findElement(By.xpath("//input[@id='user']"));
driver.findElement(By.cssSelector("#user"));
```

Q4. Program to Handle JavaScript Alert

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
```

```
public class AlertDemo {
```

```
    public static void main(String[] args) {
```

```
        WebDriver driver = new ChromeDriver();
```

```
        driver.get("https://the-internet.herokuapp.com/javascript_alerts");
```

```
        driver.findElement(By.xpath("//button[text()='Click for JS Alert']"))
            .click();
```

```

        driver.switchTo().alert().accept();

        System.out.println("Alert Accepted Successfully");

        driver.quit();
    }
}

```

Q5. Difference Between Implicit Wait and Explicit Wait

Implicit Wait	Explicit Wait
Applies to all elements	Applies to specific element
Set once	Set whenever needed
Waits for specified time	Waits until condition is met
Less flexible	More flexible

Implicit Wait Example

```

driver.manage().timeouts()
    .implicitlyWait(Duration.ofSeconds(10));

```

Explicit Wait Example

```

import org.openqa.selenium.support.ui.WebDriverWait;
import org.openqa.selenium.support.ui.ExpectedConditions;
import java.time.Duration;

```

```
WebDriverWait wait =  
    new WebDriverWait(driver, Duration.ofSeconds(10));  
  
wait.until(  
    ExpectedConditions.elementToBeClickable(  
        By.id("loginBtn")  
    )  
);  
  
driver.findElement(By.id("loginBtn")).click();
```